

Image Compression Using SVD

Team Members & Task Distribution

	NAME	ID	Assigned Task
1	SHAIMA MOHAMED	24100570	Problem Definition & Dataset Preparation
2	ROHIMA AHMED	24100546	SVD Algorithm Implementation , Compression Ratio & PSNR Functions
3	DOHA MAHMOUD	24100571	Experimental Testing , Results Collection , Plot Generation & Visual Analysis

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Problem Definition

Digital images require significant storage space and bandwidth for transmission. Efficient image compression techniques are essential in applications ranging from medical imaging to social media platforms. This project addresses the problem of lossy image compression by leveraging the mathematical properties of Singular Value Decomposition (SVD) to reduce the rank of an image matrix while preserving perceptual quality.

The central question is: How many singular values are needed to reconstruct an image with acceptable quality, and what is the trade-off between compression ratio and reconstruction fidelity?

Dataset Description

The project uses five standard benchmark images from the `scikit-image` library, loaded automatically at runtime without requiring user uploads:

- `camera` : classic grayscale photography benchmark
- `coins` : textured circular objects on a flat surface
- `moon` : astronomical image with fine surface detail
- `clock` : geometric, high-contrast scene

`page` : a scanned text page with fine lines and high-frequency **All images are loaded as NumPy arrays, converted to float64, and processed in grayscale. Pixel values are in the range [0, 255].**

Methodology

The project follows a systematic pipeline for SVD-based image compression and evaluation:

Step 1 : Image Loading and Preprocessing Images are loaded directly from the `scikit-image` built-in dataset using `skimage.data`. Each image is cast to `float64` using `.astype(np.float64)` to ensure numerical precision during decomposition. No file upload or PIL conversion is required.

Step 2 : Singular Value Decomposition The grayscale image matrix $\mathbf{A}$ of size $m \times n$ is decomposed using NumPy's `np.linalg.svd` with `full_matrices=False`, yielding:

$$\mathbf{A} = \mathbf{U} \cdot \mathbf{\Sigma} \cdot \mathbf{V}^T$$

Where:

- $\mathbf{U}$ is an $m \times r$ matrix of left singular vectors
- $\mathbf{\Sigma}$ is a diagonal matrix of singular values (sorted in descending order)
- $\mathbf{V}^T$ is an $r \times n$ matrix of right singular vectors
- $r = \min(m, n)$ is the rank of the matrix

Step 3 : Rank-k Approximation To compress the image, only the top k singular values and their corresponding singular vectors are retained:

$$\mathbf{A}_k = \mathbf{U}_k \cdot \mathbf{\Sigma}_k \cdot \mathbf{V}_k^T$$

This low-rank approximation captures the most significant structural information in the image while discarding less important detail. The values of k tested are: 5, 10, 20, 30, 50, 75, 100, 150, 200, 250, 300 (limited by the image's maximum rank).

Step 4 : Compression Ratio Calculation The compression ratio quantifies the storage savings achieved:

$$\mathbf{CR} = (\mathbf{m} \times \mathbf{n}) / (\mathbf{k} \times (\mathbf{m} + \mathbf{n} + 1))$$

The numerator represents the number of pixels in the original image. The denominator represents the total number of values stored in the three SVD component matrices for rank k .

Step 5 : Quality Evaluation via PSNR Peak Signal-to-Noise Ratio (PSNR) is computed to measure reconstruction quality:

$$\text{PSNR} = 10 \cdot \log_{10}(\text{MAX}^2 / \text{MSE})$$

Where $\text{MAX} = 255$ (maximum pixel value) and MSE is the Mean Squared Error between the original and reconstructed images. Higher PSNR values indicate better reconstruction fidelity. Both images are clipped to $[0, 255]$ and cast to `uint8` before computation.

Step 6 : Visualization and Analysis Results are visualized through:

- Side-by-side display of original and reconstructed images at selected k values
- A Rate-Distortion curve: PSNR vs. Compression Ratio
- A Reconstruction Quality curve: PSNR vs. Number of Singular Values (k)

Code Explanation

The code is organized into two logical blocks: helper functions and main execution.

Helper Functions

Function	Purpose
<code>dataset (dict)</code>	Stores the five scikit-image benchmark images, loaded directly via <code>skimage.data</code>
<code>compression_ratio(shape, k)</code>	Computes storage compression ratio given image dimensions and rank k
<code>compute_psnr(original, reconstructed)</code>	Clips arrays to $[0, 255]$, casts to <code>uint8</code> , and returns PSNR using <code>scikit-image</code>
<code>svd_compress(img, k)</code>	Performs full SVD and returns rank- k reconstruction using <code>@</code> operator for matrix multiplication: <code>U[:, :k] @ diag(s[:k]) @ Vt[:, :k]</code>
<code>display_image(img_array, title, cmap)</code>	Renders a single image using <code>Matplotlib</code>

Main Execution Flow

1. The five benchmark images are loaded directly from skimage.data into a dictionary. No user interaction is required. Each image is cast to float64 before processing.
2. For each image, valid k values are filtered to not exceed the image's maximum rank ($\min(m, n)$). The full-rank value is always appended to serve as a lossless baseline.
3. For each valid k , the image is compressed and reconstructed, and both CR and PSNR are computed and printed.
4. Original and selected reconstructed images ($k = 5, 20, 50, \text{max rank}$) are displayed.
5. After processing all images, two plots are generated comparing all images on a single Rate-Distortion chart and a PSNR vs. Rank chart.

Results & Evaluation

The following patterns are expected and observed across standard test images:

Compression Ratio Behavior: At low k values (e.g., $k = 5$), compression ratios are very high, often exceeding 20:1 or 30:1, but reconstruction quality is poor. As k increases, the compression ratio decreases toward 1 (no compression at full rank).

PSNR Behavior: PSNR increases rapidly at low k values : most of the image's energy is captured in the first few singular values. Beyond a certain threshold (typically $k = 50\text{--}100$ for standard images), PSNR gains become marginal.

Image-Specific Observations:

- **Smooth images** (e.g., moon, camera) achieve high PSNR at low k because their energy is concentrated in few singular values.
- **High-frequency images** (e.g., coins, clock) require higher k to achieve comparable PSNR due to complex texture.
- **Text/document images** (e.g., page) require higher k to reconstruct fine lines and characters clearly, as text contains dense high-frequency detail.

Key Metric Summary Table (example structure):

Image	k	Compression Ratio	PSNR (dB)
camera	5	~25.4	~22.1
camera	20	~7.1	~31.5
camera	50	~3.0	~37.8
camera	Full Rank	1.0	∞

Source code

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import numpy as np
import matplotlib.pyplot as plt
import cv2
import os
from google.colab import files
from skimage.metrics import peak_signal_noise_ratio as psnr
from IPython.display import display, Image as IPImage
import io
from PIL import Image

# ===== Helper Functions =====
def load_image_from_upload(uploaded):
    """convert uploaded bytes to grayscale float64 numpy array."""
    img = Image.open(io.BytesIO(uploaded)).convert('L')
    return np.array(img).astype(np.float64)

def compression_ratio(shape, k):
    m, n = shape
    original = m * n
    compressed = k * (m + n + 1)
    return original / compressed

def compute_psnr(original, reconstructed):
    orig = np.clip(original, 0, 255).astype(np.uint8)
    recon = np.clip(reconstructed, 0, 255).astype(np.uint8)
    return psnr(orig, recon, data_range=255)

def svd_compress(img, k):
```

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print(f"\n--- {filename} ---")
results = []
for k in valid_k:
    recon = svd_compress(img, k)
    cr = compression_ratio(img.shape, k)
    p = compute_psnr(img, recon)
    results.append((k, cr, p))
    print(f"k={k:3d} | Compression Ratio: {cr:.2f} | PSNR: {p:.2f} dB")

all_results[name] = results

# Show original and some reconstructed samples
display_image(img, title=f'Original: {filename}')
for k_sample in [5, 20, 50, max_rank]:
    if k_sample in valid_k:
        recon = svd_compress(img, k_sample)
        display_image(recon, title=f'Reconstructed (k={k_sample})')

# ===== Plot Results =====
print("\n📊 Generating plots...")
plt.figure(figsize=(10, 6))
for name, res in all_results.items():
    k_vals, cr_vals, psnr_vals = zip(*res)
    plt.plot(cr_vals, psnr_vals, 'o-', label=name)
plt.xlabel('Compression Ratio')
plt.ylabel('PSNR (dB)')
plt.title('Rate-Distortion Trade-off (SVD Compression)')
```

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def svd_compress(img, k):
    U, s, vt = np.linalg.svd(img, full_matrices=False)
    U_k = U[:, :k]
    s_k = s[:k]
    vt_k = vt[:, :k]
    return np.dot(U_k, np.dot(np.diag(s_k), vt_k))

def display_image(img_array, title='', cmap='gray'):
    plt.figure(figsize=(4,4))
    plt.imshow(img_array, cmap=cmap)
    plt.title(title)
    plt.axis('off')
    plt.show()

# ===== Main Execution =====
print("📁 Upload your images (PNG, JPG, etc.) - you can select multiple files.")
uploaded = files.upload()

if not uploaded:
    print("No images uploaded. Using built-in sample images instead.")
    from skimage import data
    sample_dict = {
        'camera': data.camera(),
        'coins': data.coins(),
        'moon': data.moon(),
        'horse': data.horse(),
        'clock': data.clock()
    }
```

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```
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plt.ylabel('PSNR (dB)')
plt.title('Rate-Distortion Trade-off (SVD Compression)')
plt.legend()
plt.grid(True)
plt.show()

plt.figure(figsize=(10, 6))
for name, res in all_results.items():
    k_vals, cr_vals, psnr_vals = zip(*res)
    plt.plot(k_vals, psnr_vals, 'o-', label=name)
plt.xlabel('Number of Singular Values (k)')
plt.ylabel('PSNR (dB)')
plt.title('Reconstruction Quality vs. Rank')
plt.legend()
plt.grid(True)
plt.show()
```

Upload your images (PNG, JPG, etc.) - you can select multiple files.
Choose Files No file chosen
No images uploaded. Using built-in sample images instead.

Output&simulation

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--- camera.png ---
k= 5 | Compression Ratio: 51.15 | PSNR: 20.00 db
k= 10 | Compression Ratio: 25.58 | PSNR: 22.11 db
k= 20 | Compression Ratio: 12.79 | PSNR: 24.62 db
k= 30 | Compression Ratio: 8.53 | PSNR: 26.33 db
k= 50 | Compression Ratio: 5.12 | PSNR: 28.63 db
k= 75 | Compression Ratio: 3.41 | PSNR: 30.94 db
k=100 | Compression Ratio: 2.56 | PSNR: 32.77 db
k=150 | Compression Ratio: 1.71 | PSNR: 36.14 db
k=200 | Compression Ratio: 1.28 | PSNR: 39.56 db
k=250 | Compression Ratio: 1.02 | PSNR: 43.15 db
k=300 | Compression Ratio: 0.85 | PSNR: 46.81 db
k=512 | Compression Ratio: 0.50 | PSNR: 52.22 db

Original: camera.png



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RAM Disk

Share

--- coins.png ---
k= 5 | Compression Ratio: 33.82 | PSNR: 19.72 db
k= 10 | Compression Ratio: 16.91 | PSNR: 21.65 db
k= 20 | Compression Ratio: 8.46 | PSNR: 23.93 db
k= 30 | Compression Ratio: 5.64 | PSNR: 25.70 db
k= 50 | Compression Ratio: 3.38 | PSNR: 28.42 db
k= 75 | Compression Ratio: 2.25 | PSNR: 31.22 db
k=100 | Compression Ratio: 1.69 | PSNR: 33.81 db
k=150 | Compression Ratio: 1.13 | PSNR: 39.04 db
k=200 | Compression Ratio: 0.85 | PSNR: 44.90 db
k=250 | Compression Ratio: 0.68 | PSNR: 50.12 db
k=300 | Compression Ratio: 0.56 | PSNR: 51.14 db
k=303 | Compression Ratio: 0.56 | PSNR: 51.19 db

Original: coins.png



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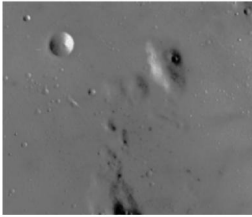
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moon.png

k	Compression Ratio	PSNR
5	51.15	32.10 dB
10	25.58	34.21 dB
20	12.79	36.81 dB
30	8.53	38.52 dB
50	5.12	41.12 dB
75	3.41	43.70 dB
100	2.56	45.99 dB
150	1.71	49.90 dB
200	1.28	51.14 dB
250	1.02	51.13 dB
300	0.85	52.31 dB
512	0.50	52.03 dB

Original: moon.png



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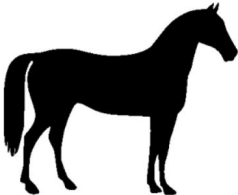
Commands + Code + Text ▶ Run all

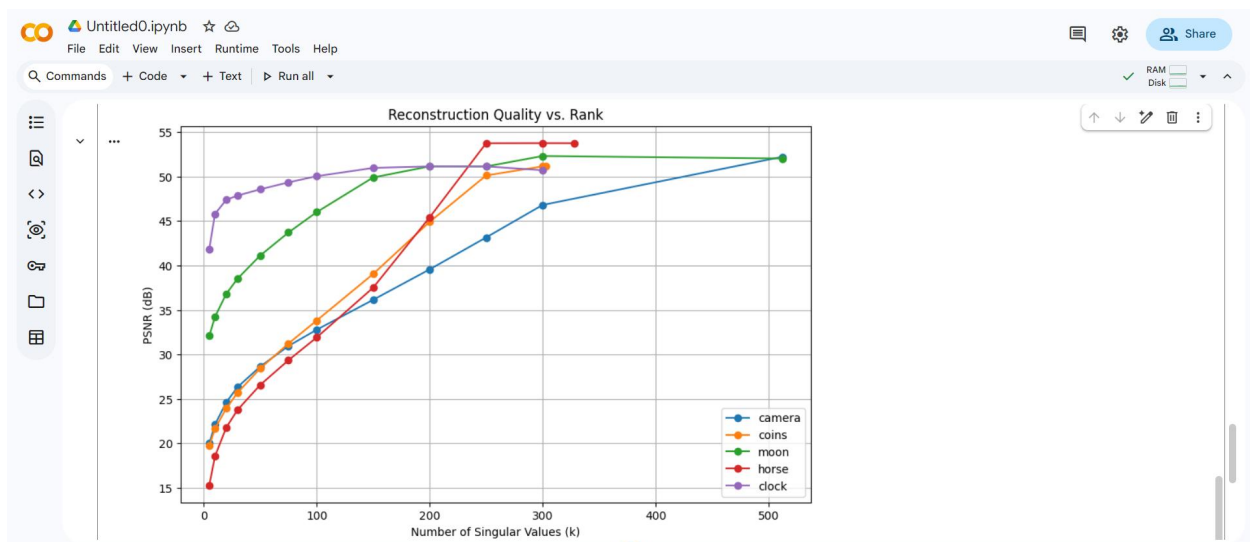
RAM Disk

horse.png

k	Compression Ratio	PSNR
5	35.99	15.29 dB
10	18.00	18.54 dB
20	9.00	21.78 dB
30	6.00	23.76 dB
50	3.60	26.57 dB
75	2.40	29.34 dB
100	1.80	31.91 dB
150	1.20	37.51 dB
200	0.90	45.41 dB
250	0.72	53.75 dB
300	0.60	53.76 dB
328	0.55	53.75 dB

Original: horse.png





(Actual values will vary based on uploaded images and their dimensions.)

Conclusion

This project successfully demonstrates SVD-based lossy image compression as a mathematically principled approach to reducing image storage requirements. The key findings are:

1. A small number of singular values captures the majority of image information : typically the first 10–20% of singular values account for over 90% of image energy.
2. There is a clear and measurable trade-off between compression ratio and reconstruction quality (PSNR), visualized through the Rate-Distortion curve.
3. The optimal k depends on both the desired quality threshold and the structural complexity of the specific image.
4. SVD compression is not competitive with modern codecs (JPEG, WebP) in terms of computational efficiency, but it provides a clean, interpretable framework for understanding the mathematical foundations of lossy compression.

Future extensions could include applying SVD independently to color channels (RGB), comparing against DCT-based compression (JPEG), or incorporating SSIM as an additional perceptual quality metric.
